Technical Report

Bilingual Text-to-Speech with FastSpeech 2: English (Svarah) and Telugu (IndicTTS)

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Assignment: Implementation of FastSpeech 2-Based Text-to-Speech Synthesis, 7 November 2026

Code and results: https://github.com/sriharshitha-19/fastspeech2-tts

Summary

We trained two separate FastSpeech 2 models, one for Indian-accented English (Svarah) and one for Telugu (IndicTTS), starting from the open-source implementation by ming024 (MIT licence). Both models were trained on about 3.7 hours of audio and validated on about 0.74 hours, on Kaggle T4 GPUs with a fixed 120-minute training budget. English stopped at step 15,629 and Telugu at about step 14,800, roughly half of the 30,000 steps that were planned.

The Telugu model produced intelligible speech for two of the five test sentences (T2a and T2b). The English model did not produce intelligible speech for any test sentence: the output is broken, noisy audio in which no words can be made out. Validation loss reached its minimum very early in both runs (step 2,500 for English, step 1,000 for Telugu) and then rose, while training loss kept falling, so both models over-fit the small training sets. The predicted duration, pitch and energy contours look plausible (declining F0, longer final symbols), so the failures appear to come mainly from the mel decoder and the training data rather than from the variance predictors. This report states these limitations openly and lists the changes that would be tried next.

1. Implementation approach and modifications

1.1 Components taken from the reference implementation

We used github.com/ming024/FastSpeech2 (an unofficial PyTorch implementation of FastSpeech 2, MIT licence) as the starting point and kept its model code unchanged: the transformer encoder (4 feed-forward-transformer layers, hidden size 256, 2 heads), the variance adaptor (duration, pitch and energy predictors with 256-bin quantised embeddings for pitch and energy, and a length regulator), the mel decoder (6 layers) with PostNet, the loss functions, the data loader and the HiFi-GAN vocoder wrapper. The vocoder is the pretrained universal HiFi-GAN generator from the same repository (no vocoder training was done). The English model has 35,598,401 trainable parameters.

1.2 Changes shared by both languages

- Own train/validation split: train and validation clips come from separate folders (raw_data/<dataset>/train and /val), so there is no overlap between them. The reference code splits one folder by a random validation size.
- preprocess.py: fix for empty silence labels in the alignment, a check for phones that are not in the symbol table (affected clips are written to skipped.txt instead of crashing), and a --limit option for a quick smoke test on a few clips.

- train.py: a wall-clock training budget (--max_minutes) that ends training cleanly with a final checkpoint, keeps only the three most recent checkpoints, and writes training and validation losses to output/log/train/log.txt and output/log/val/log.txt. The final step was set to 30,000 with validation every 500 steps and a checkpoint every 1,000 steps (the reference uses 900,000 steps).
- Compatibility fixes for current library versions: patch_librosa.py (librosa and torch.load changes), and a float32 cast of the energy targets in utils/tools.py (the first training step failed with a float64/float32 mismatch).
- New inference.py: one script for both languages that reads config.yaml, loads the latest checkpoint, synthesises a list of test sentences, saves the audio and plots the predicted duration, pitch (F0) and energy for every sentence. plot_losses.py produces the loss plots from the log files.

1.3 English-specific

English uses ARPAbet phones. At inference the text front end is the reference one: a lexicon lookup, with the rule-based g2p_en model for words that are not in the lexicon, and punctuation mapped to a short pause symbol. The configuration is multi-speaker (Svarah has many speakers), with max_seq_len raised to 1800 because some clips are long. All test audio was synthesised with speaker ID 0.

1.4 Telugu-specific

Telugu is largely phonetic, so a character-level mapping was used instead of a phone dictionary: every Telugu Unicode character is one symbol. We replaced the English and pinyin symbols in text/symbols.py with the Telugu Unicode block (U+0C00 to U+0C7F), the ASCII digits and the silence symbols. Consonant clusters are therefore represented as consonant, virama and consonant. Forced alignment for the duration targets was obtained with a Montreal Forced Aligner model trained on the Telugu data itself with a grapheme-based dictionary, since no ready-made Telugu aligner was available. The configuration is single-speaker, with max_seq_len set to 1500 because clips reach about 15 s (1,292 mel frames). At inference the Telugu front end (preprocess_telugu in inference.py) normalises the text to Unicode NFC, removes zero-width joiners, maps each Telugu character and each digit to a symbol, turns other punctuation into a pause symbol and ignores spaces.

1.5 Test sentences

The Telugu sentences in the assignment PDF were damaged by copy-and-paste (for example broken conjuncts in నిశ్శబ్ంద గా and సిద్ంధ గా). We retyped them as correct Telugu: the stray spaces inside words were removed (T3 is అడవిలో, T2b uses the compound పట్టుచీరను) and the broken conjuncts were repaired. The English sentences were kept exactly as given, including the odd spacing in "rhyth m" (E3) and the unusual word "andaroo" (E4), as the assignment asks for the text to be preserved. We note both choices here and in the repository.

2. Datasets, preprocessing and training configuration

2.1 Data used

	English (Svarah)	Telugu (IndicTTS Telugu)
Source	Svarah, Indian-accented English (multi-speaker, recorded for speech recognition, so noisier than studio TTS data)	IndicTTS Telugu (read speech); one speaker was used
Selected clips (train / val)	3,040 / 607	2,315 / 459
Clips kept after preprocessing	2,979 train (3.68 h) / 599 val (0.74 h); 69 skipped	2,314 train (3.67 h) / 459 val (0.74 h); 1 skipped
Skipped clips	Mostly 'empty alignment' and 'no TextGrid'; one with no voiced frames	One clip without a TextGrid
Audio	Resampled to 22,050 Hz, mono	Resampled to 22,050 Hz, mono
Alignments (TextGrids)	3,638 files, ARPAbet phones	Character-level, from an aligner trained on the Telugu data

The selected clips total 5.00 h for training and 1.00 h for validation in both languages. The hours in the table are lower because preprocessing measures audio after trimming leading and trailing silence (using the alignment) and after skipping clips that could not be used, so the audio the models actually saw is about 3.7 h for training and 0.74 h for validation. English: Svarah has no speaker IDs, so 115 speaker profiles were built from the metadata columns (gender, age group, language, state, district, qualification, job category, occupation domain); a profile only approximates a speaker. Clips of 1 to 20 s were kept (5,331 clips), validation was drawn first at random (seed 42) until it reached 1.00 h, identical sentences were kept on one side only, and training clips were then sampled until they reached 5.00 h. That gives 3,040 and 607 clips; the counts are simply what adds up to those hours. All 115 profiles occur in training and 108 in validation. Forced alignment produced 3,638 TextGrids (3,032 train, 606 validation). Telugu: the dataset's gender column was identified by listening (0 female, 1 male). Gender 1 was chosen because it has 8.08 h of 2 to 15 s clips against 3.74 h for gender 0, and was assumed to be a single speaker (8 clips checked by ear, all male). The same procedure gave 2,315 and 459 clips. No pretrained Telugu aligner exists, so one was trained on the data; 2,773 of 2,774 clips aligned.

2.2 Preprocessing

- Audio was exported from the Hugging Face datasets as 22.05 kHz WAV files in separate train and val folders.
- Forced alignment gives the duration of every phone (or Telugu character). preprocess.py then computes 80-bin mel spectrograms (filter length 1024, hop 256, window 1024, 0 to 8,000 Hz), pitch (F0 with pyworld, phoneme-level averages, normalised with the training statistics) and energy (phoneme-level, normalised).
- Leading and trailing silences are trimmed using the alignment. Clips with unknown phones, empty alignments or no voiced frames are skipped and listed in skipped.txt.
- Training and validation statistics (stats.json) and the speaker list (speakers.json) are saved with the preprocessed data and are needed again at inference.

2.3 Training configuration

Setting	English	Telugu
Model	FastSpeech 2 (reference architecture)	FastSpeech 2 (reference architecture)
Speakers	Multi-speaker	Single speaker
Pitch / energy	Phoneme-level, normalised	Phoneme-level, normalised
max_seq_len	1800	1500
Batch size	16 (reference value)	16
Warm-up / LR anneal	2,000 steps; ×0.3 at 20k, 25k, 28k	2,000 steps; ×0.3 at 20k, 25k, 28k
Planned steps	30,000	30,000
Steps actually run	15,629	about 14,800
Why it stopped	120-minute time budget	120-minute time budget
Approx. epochs	about 84	about 100
Validation / checkpoint interval	every 500 / 1,000 steps	every 500 / 1,000 steps
Hardware	Kaggle GPU T4 ×2	Kaggle GPU T4 ×2
Vocoder	Pretrained universal HiFi-GAN	Pretrained universal HiFi-GAN

The number of epochs is estimated as steps × 16 divided by the number of training clips.

“The values in this table are those in English/config.yaml and Telugu/config.yaml in the repository.”

3. Training and validation loss

The figures below show the total loss (left) and the individual loss terms (right) for each model. The numbers quoted in the text come from the log files.

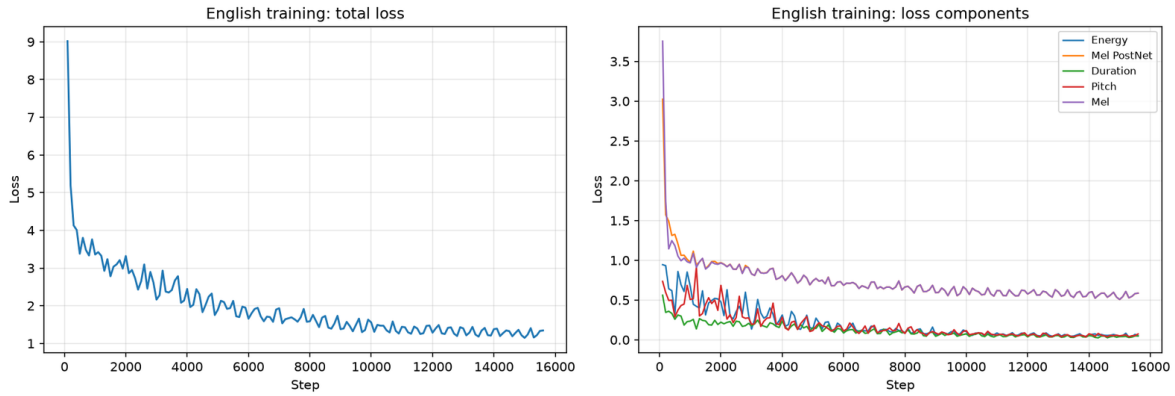


Figure 1. English: training loss.

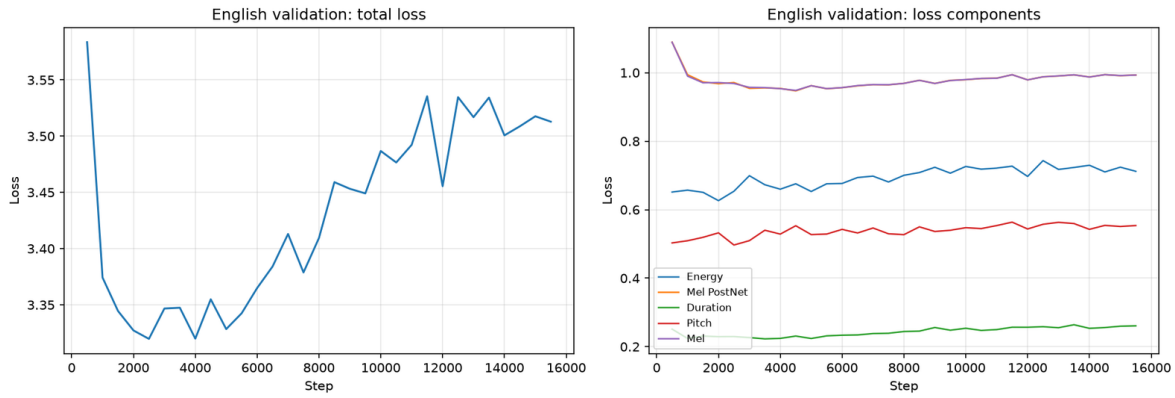


Figure 2. English: validation loss.

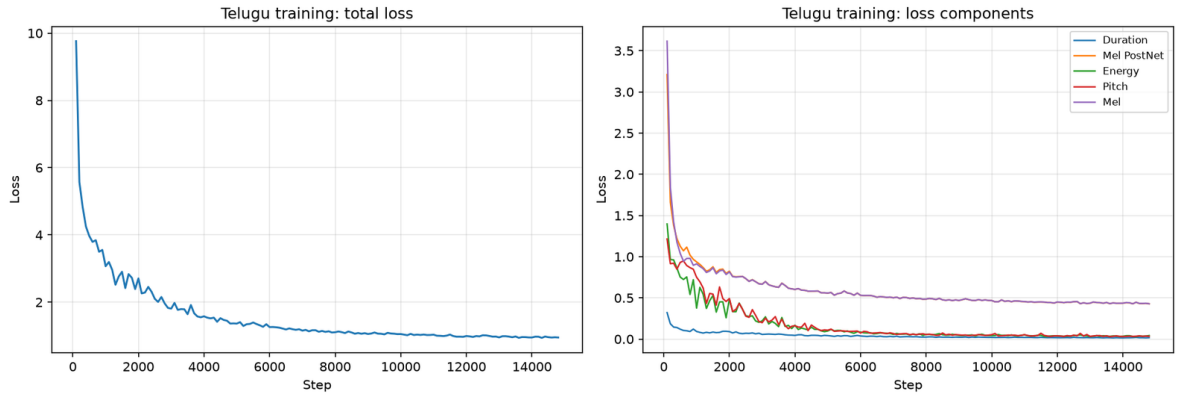


Figure 3. Telugu: training loss.

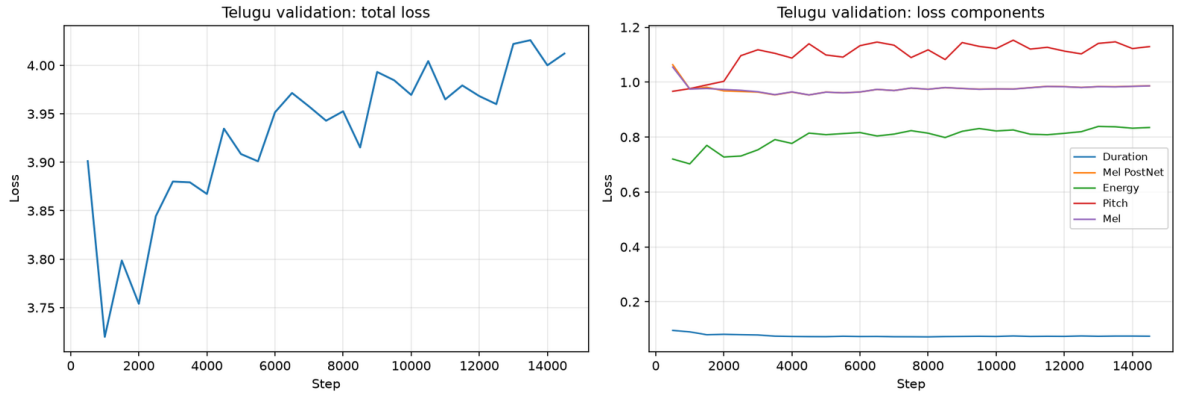


Figure 4. Telugu: validation loss.

Loss (total)	**English**	**Telugu**
Training, first logged	9.02 (step 100)	9.76 (step 100)
Training, final	1.34 (step 15,600)	0.95 (step 14,800)
Training, best	1.15 (step 15,000)	0.94 (step 14,300)
Validation, first	3.58 (step 500)	3.90 (step 500)
Validation, best	3.32 (step 2,500)	3.72 (step 1,000)
Validation, final	3.51 (step 15,500)	4.01 (step 14,500)

Convergence. In both runs the training loss falls quickly in the first 2,000 steps and then slowly. At the end the training mel loss is about 0.55 for English and 0.43 for Telugu, and the pitch, energy and duration losses are close to zero (about 0.05 or below). The English training curve has a small periodic ripple, probably related to the order in which length-grouped batches are visited; we did not investigate it.

Over-fitting. The validation loss does not follow the training loss. It reaches its minimum early (English 3.32 at step 2,500, with a similar value at step 4,000; Telugu 3.72 at step 1,000) and then rises gradually to 3.51 and 4.01. The component curves show where: the validation mel and PostNet losses stay flat at about 0.95 to 1.0 from the first few thousand steps and never improve, while the validation pitch loss (Telugu 0.97 to 1.13, English 0.50 to 0.55), energy loss (Telugu 0.72 to 0.83, English 0.65 to 0.71) and, for English, the duration loss (0.22 to 0.26) increase. With about 3.7 hours of data and roughly 85 to 100 passes over it, the networks memorise the training clips. The gap between training and validation loss is large for both languages, and larger for Telugu in the final total loss.

Instability. We saw no divergence or loss spikes. The only irregularity is the ripple mentioned above. Because the time-budgeted train.py keeps only the last three checkpoints, the early minimum-validation checkpoints were no longer available, so all test audio uses the final checkpoint (English step 15,629, Telugu about step 14,800). Saving the best-validation checkpoint would have been a better choice.

4. Observations on the generated speech

All audio files are in Results/<Language>/test_generated_audios. The listening notes below were made by the author, who listened to each file once; they are informal, and we did not run a formal listening test or compute an ASR-based error rate. One-off Whisper transcriptions of both languages are described in Sections 4.1 and 4.2.

4.1 English

ID	Test sentence	Text front end	Listening observation
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	The long word is not in the lexicon and is converted by g2p_en (85 symbols, 5.94 s).	No intelligible words; broken, noisy audio (listener note: stuck).
E2a	The orchestra's conduct during the ceremony was praised by the audience.	'conduct' is read as the noun, K AA1 N D AH0 K T (50 symbols, 3.99 s).	No intelligible words; broken noise (stuck).
E2b	The researchers will conduct an experiment to evaluate the proposed method.	'conduct' receives the same noun pronunciation, although it is a verb here (56 symbols, 3.89 s).	No intelligible words; broken noise (stuck).
E3	The rhyth m of the orchestra changed unexpectedly.	'rhyth' and the letter 'm' are processed as two tokens: R AY1 DH followed by EH1 M (37 symbols, 2.89 s).	Poor; no clear words (broken noise).

ID	Test sentence	Text front end	Listening observation
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	The numbers and times are expanded to English words by the front end, and the end of '2026' gets an odd phone sequence; 'andaroo' goes through g2p_en (79 symbols, 7.14 s).	Poor; no clear words (broken noise).

The English output is broken, crackling noise without recognisable words for all five sentences, so pronunciation and quality cannot be judged sentence by sentence. Two observations come from the text front end rather than the audio: the lexicon has one pronunciation per word, so the noun/verb pair in E2a and E2b is not distinguished, and the spacing error in E3 changes the input symbols. Neither would matter much until the model produces intelligible speech at all.

As an additional check, we transcribed the five English files with Whisper large-v3 (language set to English). None of the intended sentences was recovered. The transcripts for E1, E2a, E2b and E3 are fluent but unrelated to the input (for E2b, a repeated phrase), which is typical when a recognizer is given noise and is consistent with our listening notes. The E4 transcript shares only a few loose fragments with the input (for example ‘to meet’, ‘10th’ and a number). Because Whisper can produce fluent text from noise, these transcripts show that no intended words were recognizable but say nothing about the quality of individual sounds.

4.2 Telugu

ID	Test sentence	Text front end	Listening observation
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	A long loanword with many conjuncts (62 symbols, 4.57 s).	Poor (listener note: bad).
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	Short sentence, common words (28 symbols, 2.35 s).	Good: intelligible (listener note: good).
T2b	ఆమె పట్టుచీరను ధరించి వేడుకకు వచ్చింది.	'పట్టు' appears inside a compound (34 symbols, 2.30 s).	Good: intelligible (listener note: good).
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.	Contains the geminate and conjunct sequence in 'నిశ్శబ్దంగా' (29 symbols, 1.82 s).	Poor (listener note: bad).
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.	Digits and the time are not normalised: '2026' becomes the symbols 2 0 2 6 and '10:45' becomes 1 0, pause, 4 5 (78 symbols, 5.85 s).	Poor (listener note: bad).

The Telugu model produced intelligible speech for the two short, common-vocabulary sentences (T2a and T2b). The word 'పట్టు' means both 'grip' (T2a) and 'silk' (T2b), but the two meanings are spelled and pronounced the same, so a character-level model has no ambiguity to resolve; the difference can only show up in prosody, and we did not hear a clear difference. The three failures share a property: T1 and T3 contain long conjunct sequences that may be rarer in the training data, and T4 contains digits and a time, which the front end does not convert to words, so the model

receives symbols that it has probably seen very rarely (digits are likely to be uncommon in read-speech transcripts).

As an additional check, we transcribed the five Telugu files with Whisper large-v3 (language set to Telugu). For T2a and T2b the transcripts are close to the intended sentences: T2a recovers the first words and only the ending is wrong, and T2b has small errors in 'చీరను', 'ధరించి' and 'వచ్చింది'. For T1 and T3, only the first words are recovered (a rough rendering of the long first word in T1; 'పులి అడవిలో' in T3) and the rest is garbled. T4 is mostly garbled; apart from a rough 'ఈ సమావేశం' at the start, the transcript contains no recognisable date or time words, which is consistent with the front end not expanding digits. Whisper's own Telugu recognition is imperfect, so these transcripts support the listening notes but are not a formal error rate.

5. Variance adaptor analysis (duration, pitch and energy)

For each test sentence the plots show, from top to bottom, the predicted duration of every input symbol (in ms, from the rounded frame counts), the predicted F0 over time and the predicted energy over time. Pitch and energy are predicted per symbol (phoneme-level), which is why the contours are staircase-shaped; the vertical lines mark symbol boundaries and the symbol labels are on the top axis. Pitch and energy were de-normalised with the training statistics. The summary statistics come from inference_summary.csv.

ID	Symbols	Audio (s)	Mean symbol (ms)	Longest symbol (ms)	F0 mean / range (Hz)	Energy mean	Energy max
E1	85	5.94	69.9	290.2	240 / 194–269	27.1	59.0
E2a	50	3.99	79.9	220.6	230 / 202–256	28.9	57.5
E2b	56	3.89	69.5	139.3	228 / 193–258	22.4	49.6
E3	37	2.89	78.1	197.4	235 / 172–283	26.8	60.6
E4	79	7.14	90.4	185.8	233 / 185–298	22.3	69.7
T1	62	4.57	73.8	174.1	193 / 116–273	36.9	97.3
T2a	28	2.35	83.8	150.9	203 / 146–273	40.3	97.7
T2b	34	2.30	67.6	174.1	198 / 134–276	32.7	76.9
T3	29	1.82	62.9	116.1	187 / 142–240	31.7	67.2
T4	78	5.85	75.0	174.1	190 / 126–257	33.6	85.9

5.1 General patterns

- Pitch declination: in every sentence the predicted F0 starts high and falls towards the end (for example T2a from about 273 Hz at the start to about 148 Hz at the end), as in natural declarative speech. The Telugu contours span a wider range (about 116 to 276 Hz) and the English ones a narrower, higher range (about 172 to 298 Hz).
- Final lengthening: the longest or nearly longest symbols are near the end of the sentence (E2a final 'S' about 220 ms, E3 final 'IY0' about 197 ms, E4 final symbols about 185 ms), a normal phrase-final effect.
- Duration range: most symbols are predicted between about 25 and 150 ms. The values come in steps because they are rounded frame counts (one frame is 11.6 ms). Some symbols get very short or zero durations, which means they are effectively deleted (for example a short unstressed vowel in E1).
- Energy follows the stressed syllables and falls towards the end, with large symbol-to-symbol jumps because it is predicted per symbol; Telugu energy values are higher on average than English ones.
- The contours of the failed sentences look similar to those of the successful ones. This suggests that the variance predictors learned plausible prosody, and the poor audio comes mainly from the mel decoder (whose validation loss never improved) and, for English, from the noisy training data.

5.2 Per-sentence notes and plots

Each figure is for one sentence: duration (top), pitch (middle), energy (bottom).

E1. The longest sentence in English (85 symbols). F0 starts near 270 Hz and falls in steps to about 215 Hz. Duration has a heavy right tail, with the longest symbol (about 290 ms) in the final words, and at least one symbol is predicted with zero length. Energy swings between about 7 and 59 from symbol to symbol.

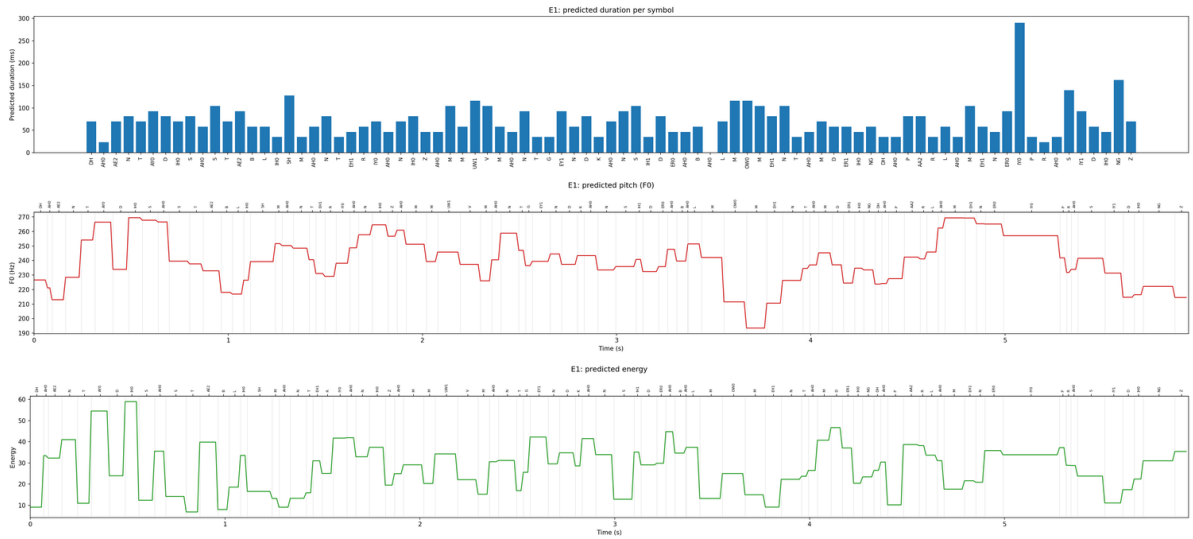


Figure: predicted duration, pitch and energy for E1.

E2a. Pitch peaks of about 240 to 256 Hz alternate with dips to about 205 Hz and end on a low plateau near 225 Hz. Durations are fairly even (about 35 to 160 ms) except for the very long final 'S' (about 220 ms). The word 'conduct' is the noun reading in both sentences.

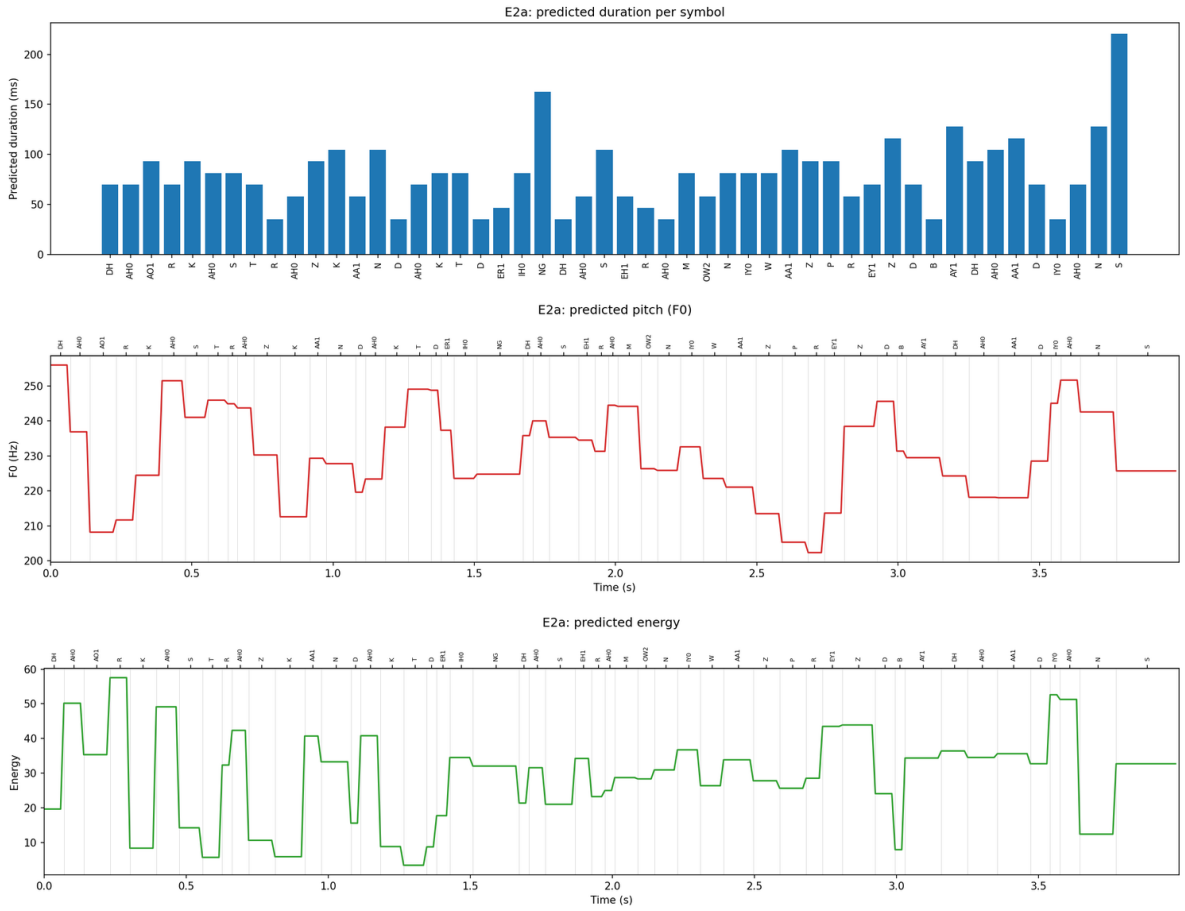


Figure: predicted duration, pitch and energy for E2a.

E2b. A high start (about 258 Hz) falls to about 195 Hz after one second and rises again in the second half. A stressed vowel just after 'conduct' is the longest symbol (about 140 ms). The contour is similar to E2a, so the different part of speech of 'conduct' is not reflected in the prosody.

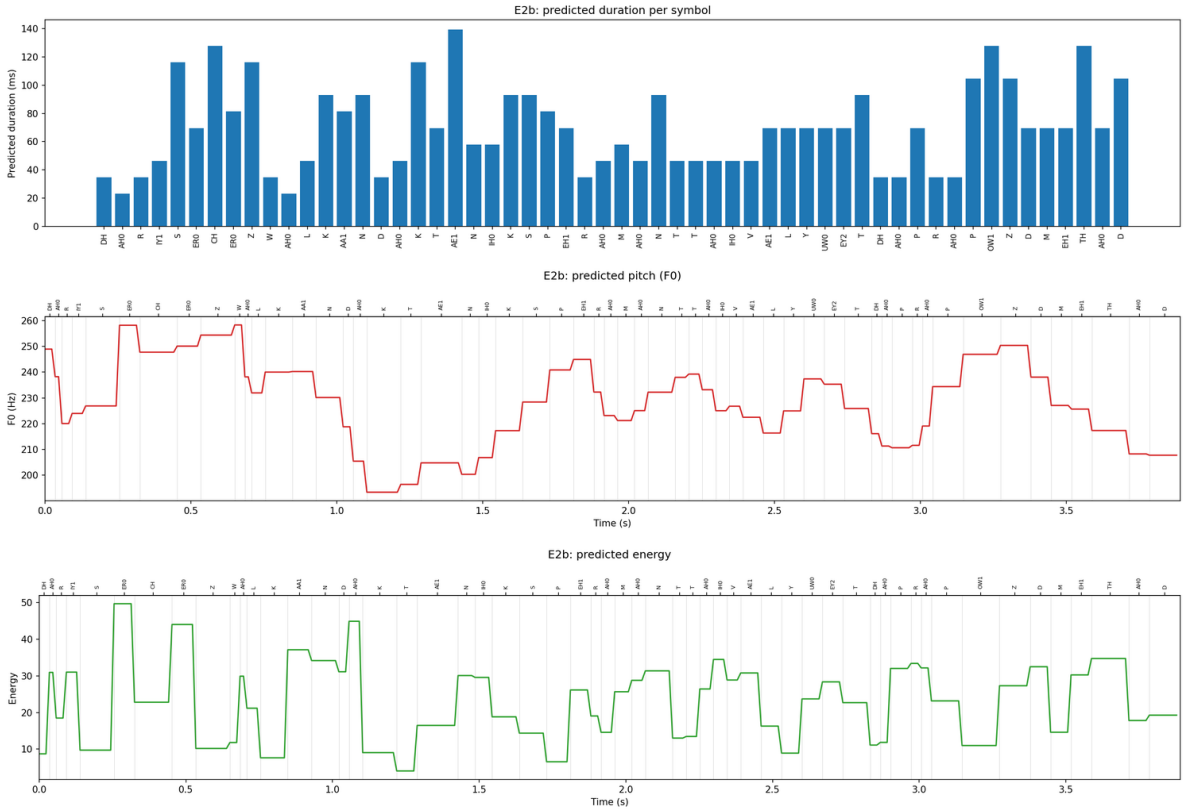


Figure: predicted duration, pitch and energy for E2b.

E3. The widest pitch swing in English (172 to 283 Hz): a peak around the first stressed syllables, a gradual fall to about 220 Hz, a dip to about 172 Hz near the end and a small rise on the last symbol. The final 'IY0' is the longest symbol (about 197 ms).

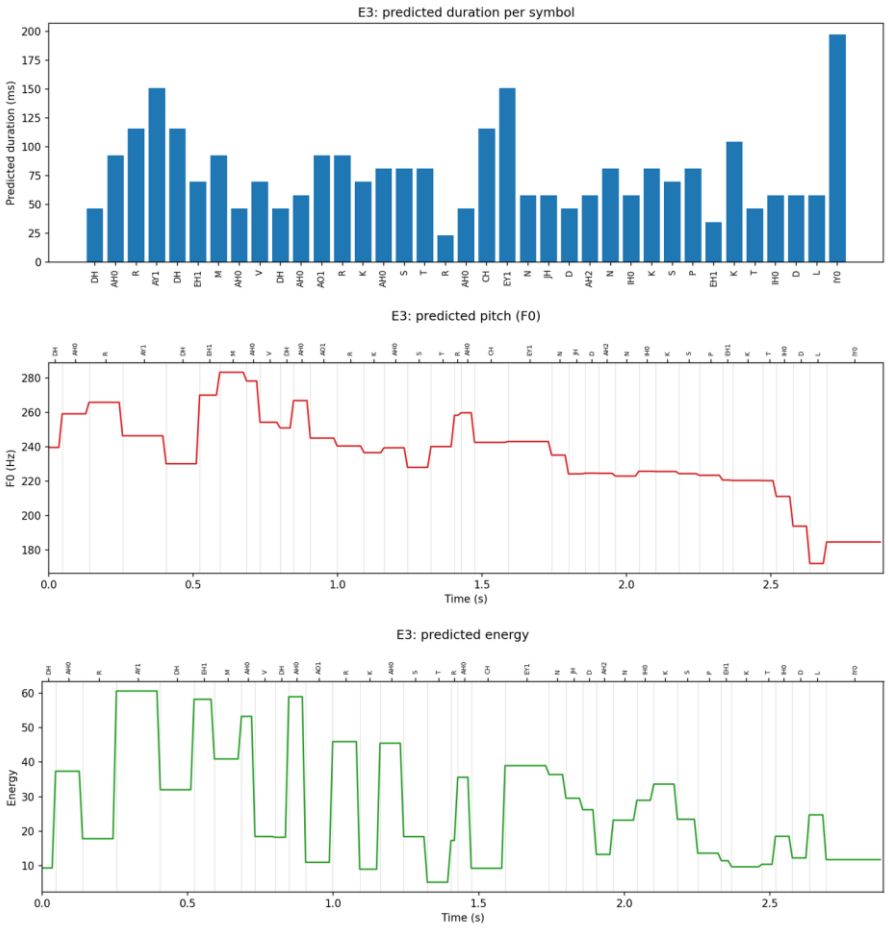


Figure: predicted duration, pitch and energy for E3.

E4. The longest English sentence (7.14 s). F0 starts at about 270 to 297 Hz and falls to about 185 Hz by 4.5 s. Durations are mostly 60 to 150 ms with several long vowels near the end. The number words from '10:45' and '2026' add many symbols and contribute to the long duration.

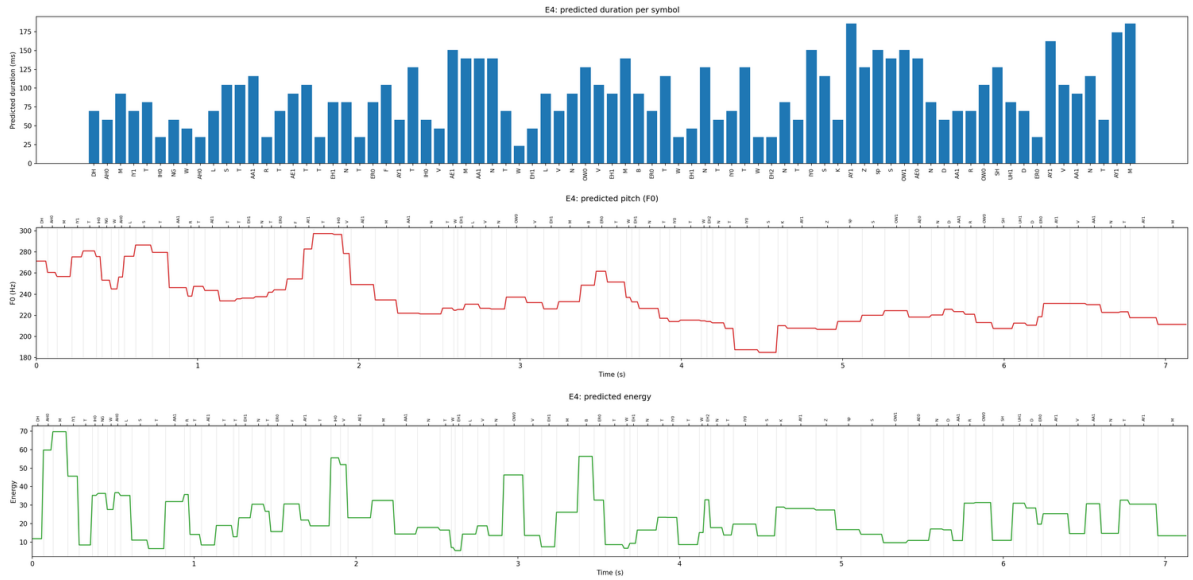


Figure: predicted duration, pitch and energy for E4.

T1. F0 rises quickly to the maximum of 273 Hz at the start, then declines in steps with a large drop to about 120 to 130 Hz around 2.8 s and a partial recovery. Durations range from about 12 to 174 ms. The long technical word is spread over many short symbols.

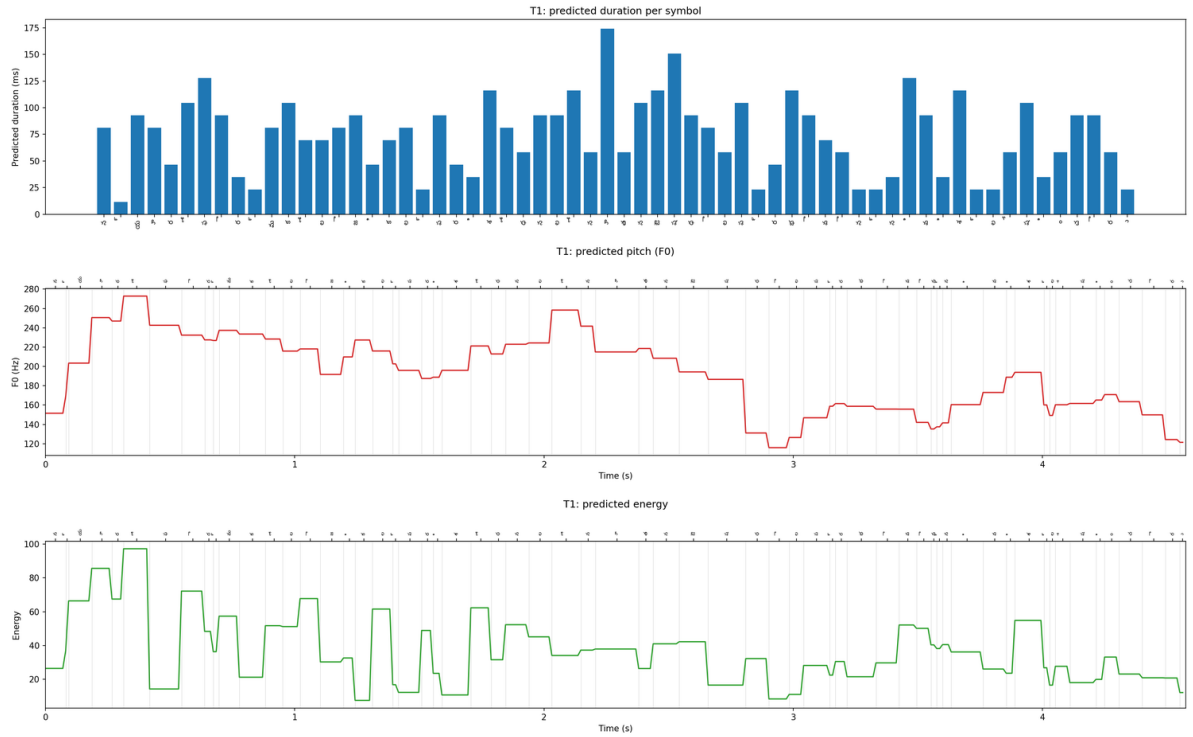


Figure: predicted duration, pitch and energy for T1.

T2a. A clean declination from about 273 Hz at the start to 148 Hz at the end, with small local peaks. Durations are 23 to 151 ms. Energy is highest on the first syllable (about 98) and then varies

between about 13 and 77, with one near-zero dip around 1.2 s. The overall pattern is regular, which fits the good listening result.

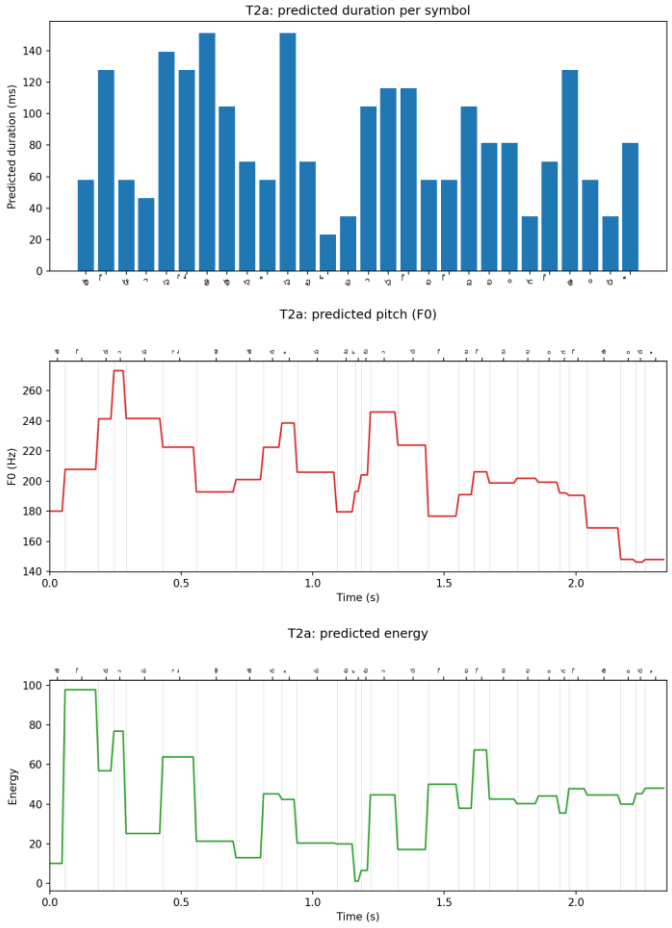


Figure: predicted duration, pitch and energy for T2a.

T2b. F0 starts low (about 140 Hz), peaks at 276 Hz in the first second and falls to about 135 to 140 Hz at the end. The first symbol is the longest (about 174 ms). The contour is similar to T2a, which is also intelligible.

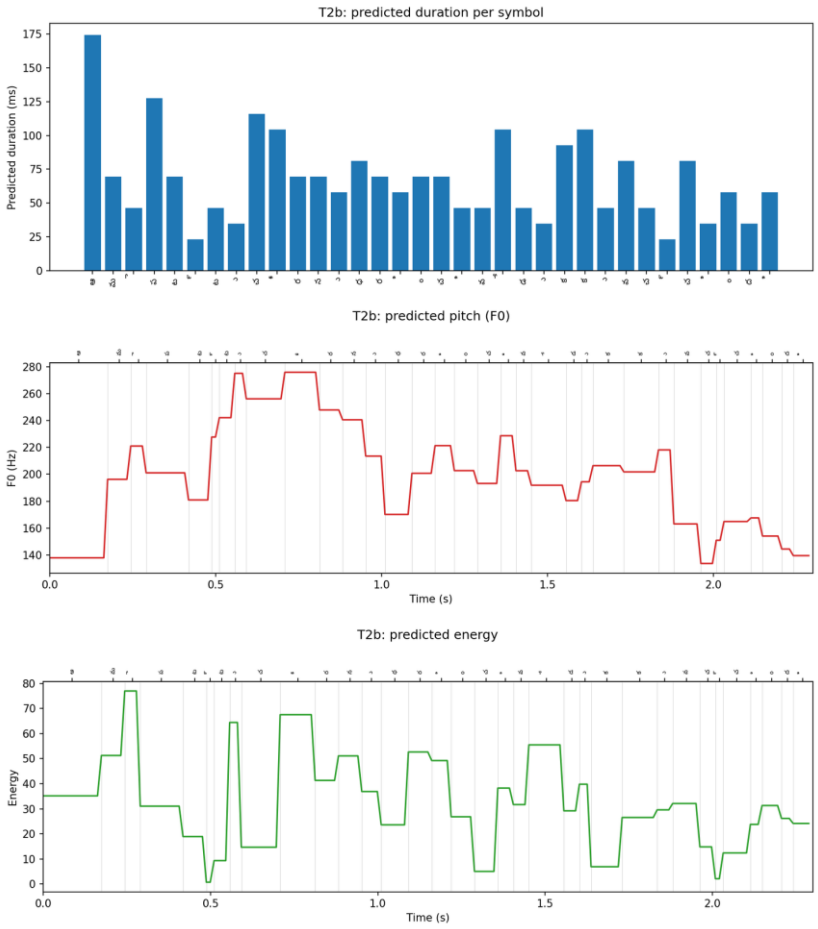


Figure: predicted duration, pitch and energy for T2b.

T3. The shortest sentence (1.82 s). F0 jumps sharply between about 150 and 240 Hz, with an early peak of 240 Hz followed by a reset to about 172 Hz, which is a less smooth contour than T2a. One

symbol is predicted with only about 12 ms. The geminate cluster in 'నిశ్శబ్దంగా' is spread over several short symbols.

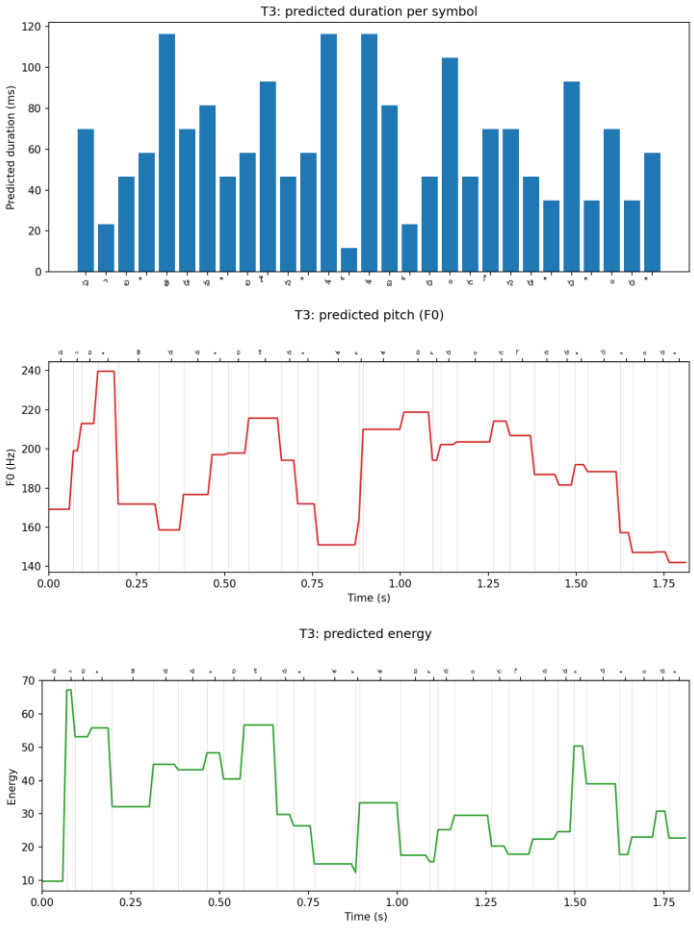


Figure: predicted duration, pitch and energy for T3.

T4. F0 declines from about 260 Hz to about 126 Hz, then rises to about 243 Hz shortly before the end and drops to about 180 Hz on the last symbols, a late rise that is not typical for a statement. Each digit symbol gets an ordinary duration (about 46 to 116 ms), so the date and time are not treated as

numbers. Energy falls close to zero twice (around 3.5 s and between 5.1 and 5.4 s), that is, the model predicts nearly silent stretches inside the sentence; we did not align these dips to individual symbols.

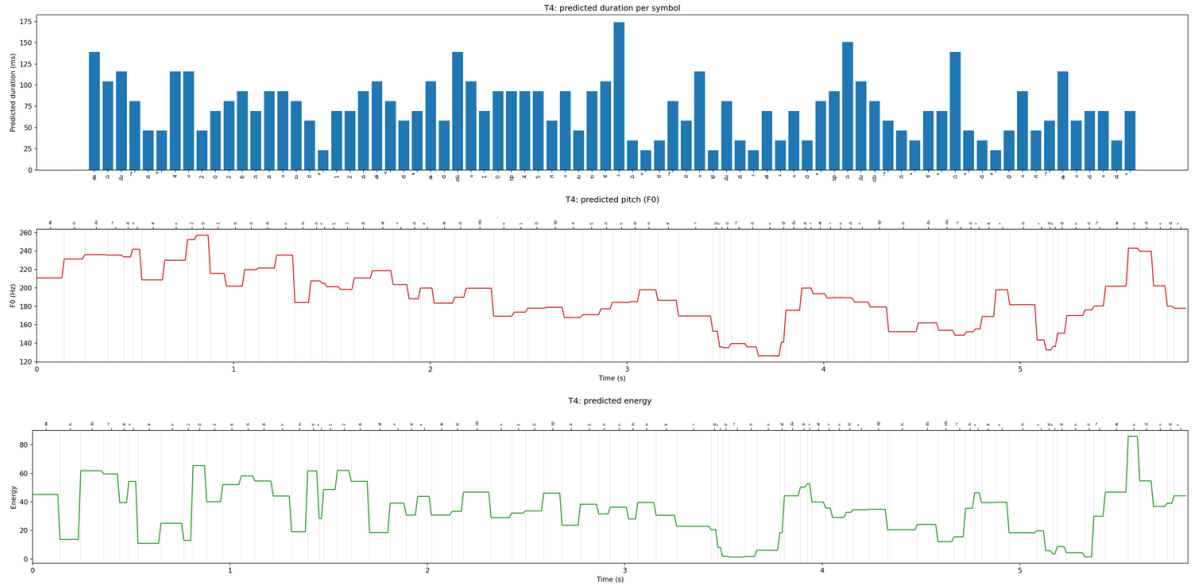


Figure: predicted duration, pitch and energy for T4.

6. Comparison of English and Telugu, limitations and improvements

6.1 Comparison

	English	**Telugu**
Result on the test sentences	No intelligible output in any of the 5 sentences	2 of 5 intelligible (T2a, T2b), 3 poor
Data	Multi-speaker, noisy, accented speech recorded for speech recognition	One speaker, read speech
Text front end	Lexicon plus g2p_en (rule-based fallback), one pronunciation per word	Character-level mapping, no dictionary needed
Best validation loss	3.32 (step 2,500)	3.72 (step 1,000)
Final validation loss	3.51	4.01
Main difficulty	Heterogeneous training audio, homographs, number expansion	Long conjuncts, digits and dates, rare characters

The Telugu model did better mainly because its training data is a single clean speaker and its character-level symbols are close to the sounds. The English data is far more varied: with several accented speakers, background noise and only 3.7 hours in total, there is limited data per speaker, and the alignments (3,638 TextGrids for 3,647 clips) may be less reliable on noisy speech. Both models are far from converged in the sense of the original paper, which trains for hundreds of thousands of steps; we trained for about 15,000. The comparison should therefore be read as a comparison of two heavily limited runs, not of the languages themselves. We could not run ablations, so the explanations above are hypotheses.

6.2 Limitations and failures

- Training data below the requested size (3.7 h instead of about 5 h) and training stopped at about half of the planned steps because of the time budget.
- Over-fitting from the first few thousand steps and no best-validation checkpoint (only the last three checkpoints were kept).
- English output is not intelligible. We did not diagnose whether the noise comes from the mel decoder, from the pretrained universal vocoder on out-of-distribution mels, or from the noisy data.
- No number, date or time normalisation for Telugu, and no homograph handling for English (E2a/E2b).
- Evaluation was informal: one listener and a single automatic transcription check (Whisper large-v3) on the English and Telugu files; no mean opinion score and no character error rate.

6.3 Possible improvements

- English: use one clean speaker (or a few speakers) selected by signal-to-noise ratio, denoise the audio, and use all the available hours of that speaker.
- Train much longer (at least 100,000 steps), with regularisation (dropout, data augmentation), a validation-based checkpoint save and early stopping.
- Start from a pretrained model (for example trained on LJSpeech or on Indian-language data) and fine-tune, which usually needs far less data.
- Text front end: Telugu number, date and time expansion to words; part-of-speech aware pronunciation for English homographs such as 'conduct'; normalise typos like 'rhyth m' only when allowed.
- Check alignment quality (skip or re-align clips with unusual phone durations) and consider a vocoder fine-tuned on the target speaker.
- Evaluate with a listening test with several listeners and compute an ASR character error rate on all outputs, instead of one informal checkA.

7. Reproducibility

7.1 Software

Python 3 with PyTorch, NumPy, SciPy, librosa (patched through patch_librosa.py), pyworld, g2p_en (with the NLTK data it needs), PyYAML, matplotlib, tgt, tqdm and soundfile. The exact list is in requirements.txt; install it with: pip install -r requirements.txt. The training and inference runs were done on Kaggle (Python 3.13, a recent PyTorch release, GPU T4 ×2); inference also runs on CPU.

7.2 Files and configuration

English/ and Telugu/ each contain train.py, preprocess.py, inference.py, config.yaml (preprocessing, model and training settings in one file) and checkpoints/. The pretrained HiFi-GAN generator (generator_universal.pth.tar) is placed in <Language>/hifigan/. The test sentences are in <Language>/test_sentences.txt (format ID|text). Results are in Results/<Language>/.

7.3 Commands (run inside the language folder, shown for English)

Data preparation: export the audio to 22,050 Hz WAV files in raw_data/<dataset>/train/ and /val/, and place the forced-alignment TextGrids in preprocessed_data/<dataset>/TextGrid/. A Kaggle notebook (fs2_kaggle.ipynb) automates these steps together with the commands below.

```
cd English
python preprocess.py config.yaml                # mel, pitch, energy, durations,
stats.json
python preprocess.py config.yaml --limit 20      # optional quick smoke test
python train.py -c config.yaml --max_minutes 120 # time-budgeted training
python inference.py --config config.yaml --sentences test_sentences.txt \
    --out_dir ../Results/English --speaker_id 0
cd ..
python plot_losses.py --train <log_dir>/train/log.txt --val <log_dir>/val/log.txt
\
    --out Results/English --name English
```

For Telugu run the same commands inside Telugu/ (no --speaker_id is needed) and use --out_dir ../Results/Telugu and --name Telugu. inference.py needs the checkpoint in checkpoints/<step>.pth.tar and stats.json in the preprocessed_data folder named in config.yaml.

7.4 Checkpoints

The trained model files (about 138 MB each, weights only) are too large for a normal Git repository, so they are provided as external download links in the README:

English Checkpoint : https://drive.google.com/file/d/1cON9Bd_YhnwyvuGb-YEWrNWEIMqmGCwJ/view?usp=sharing
Telugu Checkpoint : https://drive.google.com/file/d/1refaq0tPguNQBjVEZFJv6JqwUjvdxNZT/view?usp=sharing

7.5 Acknowledgements

The model code is adapted from github.com/ming024/FastSpeech2 (MIT licence). The English and Telugu data come from Svarah and IndicTTS Telugu, and the vocoder is the universal HiFi-GAN generator distributed with the reference repository.